

Executive Workforce Analytics Dashboard

Problem Statement & Project Documentation

Project Overview

Organizations face increasing challenges in managing employee retention, workforce productivity, job satisfaction, work-life balance, training effectiveness, and overall workforce performance. High employee attrition leads to increased recruitment costs, loss of organizational knowledge, reduced productivity, and operational disruption. HR leaders require data-driven insights to identify workforce risks and make informed decisions.

Problem Statement

The organization lacks a centralized analytics solution to monitor workforce health, employee attrition trends, productivity indicators, and risk factors contributing to employee turnover. Management needs an interactive dashboard that transforms raw employee data into actionable business insights, enabling proactive workforce planning and retention strategies.

Project Objectives Analyze workforce demographics and organizational distribution. Measure employee attrition count and attrition rate. Identify factors influencing employee turnover. Evaluate job satisfaction and work-life balance indicators. Assess the impact of overtime on employee retention. Analyze promotion delays and workforce risk. Measure training effectiveness and productivity metrics. Provide executive-level insights through interactive dashboards.

Business Questions Addressed What is the current attrition rate? Which departments contribute most to workforce composition? How does overtime influence employee turnover? Does job satisfaction impact attrition behavior? Which employees can be classified as high-risk? How do training hours relate to workforce productivity? How do performance ratings vary across the workforce? Which job roles demonstrate higher workloads?

Dashboard Structure

Page 1 – Executive Workforce Overview

Provides a high-level summary of workforce metrics including employee count, attrition rate, average income, tenure, workforce distribution, and executive insights.

Page 2 – Workforce Risk Analysis

Examines overtime impact, job satisfaction impact, promotion-related risks, and identifies workforce segments with elevated attrition probability.

Page 3 – Performance & Productivity

Focuses on performance ratings, training hours, absenteeism, project allocation, and workforce productivity indicators.

Technology Stack SQL – Data extraction and business query analysis. Python (Pandas, Matplotlib) – Data cleaning and exploratory analysis. Power BI – Dashboard development and visualization. CSV Dataset – Employee workforce dataset containing 10,000 records.

Expected Business Outcome

The solution enables HR managers and business leaders to monitor workforce performance, identify attrition risks, improve retention strategies, optimize training investments, and support data-driven decision-making. The dashboard transforms workforce data into strategic insights that improve organizational effectiveness and employee engagement.